

Importance Sampling

Monte Carlo Statistical Methods

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1 Goal and Setup

In the previous note, we introduced *importance weights* using the two-dice example: we only sample from a convenient distribution $g(x)$ but want our samples to behave as if they had come from a more complicated target distribution $f(x)$. The weights

$$w(x) = \frac{f(x)}{g(x)}$$

allow us to re-interpret samples from g as if they were drawn from f .

In this document, we go one level deeper and ask a more precise question:

Assume we can sample from both f and g . For a given integrand $h(x)$, is it better to use plain Monte Carlo (sampling from f) or importance sampling (sampling from g and using weights)?

Concretely, we want to

- define both estimators of

$$\mu = \mathbb{E}_f[h(X)] = \int h(x)f(x) dx,$$

- derive their expectations and variances carefully,
- observe that both are unbiased (zero bias),
- compare their variances and obtain a clean condition for when IS helps or hurts,
- interpret this condition in terms of *where g oversamples or undersamples relative to f , weighted by $h^2(x)$* ,
- illustrate the behavior with rare-event and non-rare-event examples.

Throughout the note:

- $f(x)$ is the **target** density (or pmf),
- $g(x)$ is a fixed **proposal** density (or pmf),
- $h : \mathcal{X} \rightarrow \mathbb{R}$ is the **integrand** of interest,
- X denotes a random variable; we distinguish f and g using $\mathbb{E}_f[\cdot]$ and $\mathbb{E}_g[\cdot]$.

Our guiding question for this document is:

Both estimators are unbiased. Which one has smaller variance, and why?

This will give a precise mathematical version of the intuition: *importance sampling only helps if the proposal puts probability mass where $h^2(x)$ is large.*

2 Plain Monte Carlo Estimator

We start with the baseline. Assume we can sample directly from $f(x)$. We draw i.i.d. samples

$$X_1, X_2, \dots, X_n \sim f(x).$$

2.1 Definition

The plain Monte Carlo estimator of

$$\mu = \mathbb{E}_f[h(X)]$$

is the sample mean

$$\hat{\mu}_{MC} = \frac{1}{n} \sum_{i=1}^n h(X_i).$$

2.2 Expectation (unbiasedness)

We compute the expectation under f :

$$\begin{aligned} \mathbb{E}_f[\hat{\mu}_{MC}] &= \mathbb{E}_f \left[\frac{1}{n} \sum_{i=1}^n h(X_i) \right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_f[h(X_i)] \quad (\text{linearity of expectation}) \\ &= \frac{1}{n} \cdot n \mathbb{E}_f[h(X)] \quad (\text{all } X_i \text{ are identically distributed as } X) \\ &= \mathbb{E}_f[h(X)] = \mu. \end{aligned}$$

Thus $\hat{\mu}_{MC}$ is **unbiased**:

$$\mathbb{E}_f[\hat{\mu}_{MC}] - \mu = 0.$$

2.3 Variance

Next, compute the variance:

$$\text{Var}_f(\hat{\mu}_{MC}) = \text{Var}_f \left(\frac{1}{n} \sum_{i=1}^n h(X_i) \right).$$

Because $X_1, \dots, X_n$ are i.i.d., the $h(X_i)$ are also i.i.d. Hence:

$$\begin{aligned} \text{Var}_f(\hat{\mu}_{MC}) &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}_f(h(X_i)) \quad (\text{independence, identical variance}) \\ &= \frac{1}{n^2} \cdot n \text{Var}_f(h(X)) \\ &= \frac{1}{n} \text{Var}_f(h(X)). \end{aligned}$$

Writing the variance in terms of second moments:

$$\text{Var}_f(h(X)) = \mathbb{E}_f[h^2(X)] - \mu^2,$$

we obtain

$$\text{Var}_f(\hat{\mu}_{MC}) = \frac{1}{n} \left(\mathbb{E}_f[h^2(X)] - \mu^2 \right).$$

Interpretation. The variance of the Monte Carlo estimator decreases as $1/n$, but the constant in front,

$$\mathbb{E}_f[h^2(X)] - \mu^2,$$

depends on *how wild* $h(X)$ is under f . If $h^2(X)$ is large on regions where f puts substantial mass, then plain MC has high variance.

3 Importance Sampling Estimator

Now we introduce importance sampling. Assume $g(x)$ is another density such that

$$f(x) > 0 \implies g(x) > 0.$$

We define the importance weight

$$w(x) = \frac{f(x)}{g(x)}.$$

3.1 Integral identity

Recall the identity:

$$\begin{aligned} \mu &= \mathbb{E}_f[h(X)] = \int h(x)f(x) dx \\ &= \int h(x) \frac{f(x)}{g(x)} g(x) dx = \mathbb{E}_g[h(X)w(X)]. \end{aligned}$$

This says: if $X \sim g(x)$, then $h(X)w(X)$ has the correct expectation μ .

3.2 Definition of the IS estimator

We now draw i.i.d. samples from the proposal:

$$X_1, X_2, \dots, X_n \sim g(x),$$

and form the importance sampling estimator:

$$\hat{\mu}_{IS} = \frac{1}{n} \sum_{i=1}^n h(X_i)w(X_i), \quad w(X_i) = \frac{f(X_i)}{g(X_i)}.$$

3.3 Expectation (unbiasedness)

Compute its expectation under g :

$$\begin{aligned} \mathbb{E}_g[\hat{\mu}_{IS}] &= \mathbb{E}_g \left[\frac{1}{n} \sum_{i=1}^n h(X_i)w(X_i) \right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_g[h(X_i)w(X_i)] \quad (\text{linearity of expectation}) \\ &= \frac{1}{n} \cdot n \mathbb{E}_g[h(X)w(X)] \quad (\text{all } X_i \text{ are i.i.d. as } X \sim g) \\ &= \mathbb{E}_g[h(X)w(X)] = \mu. \end{aligned}$$

Thus $\hat{\mu}_{IS}$ is also **unbiased**:

$$\mathbb{E}_g[\hat{\mu}_{IS}] - \mu = 0.$$

Key point. Both estimators have exactly the same bias (zero). When we compare them, we are really comparing their *variances*, i.e., their mean squared errors.

3.4 Variance under g

We now compute the variance of $\hat{\mu}_{IS}$ under g :

$$\text{Var}_g(\hat{\mu}_{IS}) = \text{Var}_g\left(\frac{1}{n} \sum_{i=1}^n h(X_i)w(X_i)\right).$$

By independence and identical distribution of $h(X_i)w(X_i)$:

$$\begin{aligned} \text{Var}_g(\hat{\mu}_{IS}) &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}_g(h(X_i)w(X_i)) \\ &= \frac{1}{n^2} \cdot n \text{Var}_g(h(X)w(X)) \\ &= \frac{1}{n} \text{Var}_g(h(X)w(X)). \end{aligned}$$

Expanding the variance:

$$\begin{aligned} \text{Var}_g(h(X)w(X)) &= \mathbb{E}_g[h^2(X)w^2(X)] - (\mathbb{E}_g[h(X)w(X)])^2 \\ &= \mathbb{E}_g[h^2(X)w^2(X)] - \mu^2. \end{aligned}$$

Thus

$$\text{Var}_g(\hat{\mu}_{IS}) = \frac{1}{n} \left(\mathbb{E}_g[h^2(X)w^2(X)] - \mu^2 \right).$$

3.5 Rewriting IS variance under f

To compare $\text{Var}_g(\hat{\mu}_{IS})$ with $\text{Var}_f(\hat{\mu}_{MC})$, it is convenient to express everything as expectations with respect to f .

We use the fact that

$$\mathbb{E}_g[h^2(X)w^2(X)] = \int h^2(x) \frac{f^2(x)}{g(x)} dx.$$

But

$$\frac{f^2(x)}{g(x)} = f(x) \frac{f(x)}{g(x)} = f(x)w(x).$$

Therefore

$$\begin{aligned} \mathbb{E}_g[h^2(X)w^2(X)] &= \int h^2(x) w(x) f(x) dx \\ &= \mathbb{E}_f[h^2(X)w(X)]. \end{aligned}$$

Substituting back:

$$\text{Var}_g(\hat{\mu}_{IS}) = \frac{1}{n} \left(\mathbb{E}_f[h^2(X)w(X)] - \mu^2 \right).$$

Interpretation. Compared with plain MC, importance sampling replaces $\mathbb{E}_f[h^2(X)]$ by $\mathbb{E}_f[h^2(X)w(X)]$ in the variance formula.

4 Direct Variance Comparison

We now place the two variances side by side, *both* written under f .

Plain Monte Carlo variance:

$$\text{Var}_f(\hat{\mu}_{MC}) = \frac{1}{n} \left(\mathbb{E}_f[h^2(X)] - \mu^2 \right).$$

Importance sampling variance:

$$\text{Var}_g(\hat{\mu}_{IS}) = \frac{1}{n} \left(\mathbb{E}_f[h^2(X)w(X)] - \mu^2 \right).$$

Subtracting, we obtain:

$$\begin{aligned} \text{Var}_g(\hat{\mu}_{IS}) - \text{Var}_f(\hat{\mu}_{MC}) &= \frac{1}{n} \left(\mathbb{E}_f[h^2(X)w(X)] - \mu^2 - (\mathbb{E}_f[h^2(X)] - \mu^2) \right) \\ &= \frac{1}{n} \left(\mathbb{E}_f[h^2(X)w(X)] - \mathbb{E}_f[h^2(X)] \right) \\ &= \frac{1}{n} \mathbb{E}_f[h^2(X)(w(X) - 1)]. \end{aligned}$$

Define

$$\Delta(h; f, g) = \mathbb{E}_f[h^2(X)(w(X) - 1)].$$

Then

$$\text{Var}_g(\hat{\mu}_{IS}) - \text{Var}_f(\hat{\mu}_{MC}) = \frac{1}{n} \Delta(h; f, g).$$

Since both estimators are unbiased, the one with smaller variance also has smaller MSE. Thus:

- If $\Delta(h; f, g) < 0$, then IS has *smaller* variance than plain MC.
- If $\Delta(h; f, g) > 0$, then IS has *larger* variance than plain MC.
- If $\Delta(h; f, g) = 0$, they have the same variance.

Everything therefore reduces to understanding the sign of

$$\mathbb{E}_f[h^2(X)(w(X) - 1)].$$

5 Weight–Region Interpretation

Recall

$$w(x) = \frac{f(x)}{g(x)}.$$

This compares f and g pointwise:

- If $w(x) > 1$, then $f(x) > g(x)$, so $g(x)$ **undersamples** that region compared to f .
- If $w(x) < 1$, then $f(x) < g(x)$, so $g(x)$ **oversamples** that region compared to f .

The quantity

$$\Delta(h; f, g) = \mathbb{E}_f[h^2(X)(w(X) - 1)] = \int h^2(x)(w(x) - 1)f(x) dx$$

is an f -weighted average of $w(x) - 1$, with weights $h^2(x)$.

Where $h^2(x)$ is large, the value of $w(x) - 1$ matters a lot. Where $h^2(x)$ is small, the contribution is negligible.

This gives a clean interpretation:

- If $h^2(x)$ is mostly large in regions where $w(x) < 1$ (i.e. $g(x) > f(x)$, proposal oversamples the important region), then $h^2(x)(w(x) - 1)$ is mostly negative, and $\Delta(h; f, g) < 0$. **IS reduces variance.**
- If $h^2(x)$ is mostly large in regions where $w(x) > 1$ (i.e. $g(x) < f(x)$, proposal undersamples the important region), then $h^2(x)(w(x) - 1)$ is mostly positive, and $\Delta(h; f, g) > 0$. **IS increases variance.**

In words:

Importance sampling only helps if the proposal g oversamples exactly those regions of the space where $h^2(x)$ is large.

If g oversamples the “wrong” regions (where $h^2(x)$ is small) or undersamples the important regions, then the weights $w(x)$ amplify noise and the IS estimator becomes worse than plain MC.

6 Graphical Intuition (Placeholders)

Later we will insert plots that visualize $f(x)$, $g(x)$, $h(x)$ and $w(x)$ on the same axis.



Figure placeholder. Top-left: fixed target $f(x)$ and proposal $g(x)$. Top-right: a function $h_1(x)$ with $h_1^2(x)$ large where $w(x) < 1$ (IS helps). Bottom-left: a function $h_2(x)$ with $h_2^2(x)$ large where $w(x) > 1$ (IS hurts). Bottom-right: a mixed case $h_3(x)$.

These figures will visually show: - where f and g put mass, - where $h^2(x)$ is large, - and how $w(x)$ (the ratio f/g) behaves in those regions.

7 Examples: When IS Helps and When It Hurts

We now fix f and g and let $h(x)$ vary. This isolates the effect of the integrand on the variance comparison.

7.1 Example 1: Indicator of a rare event (good proposal)

Let A be a rare event under f (for example, a tail event). Define

$$h(x) = \mathbb{1}_A(x), \quad h^2(x) = \mathbb{1}_A(x).$$

Then

$$\begin{aligned}\Delta(h; f, g) &= \mathbb{E}_f[\mathbb{1}_A(X) (w(X) - 1)] \\ &= \int_A (w(x) - 1) f(x) dx.\end{aligned}$$

Notice: - On A we have $h^2(x) = 1$, so the contribution to Δ comes only from A . - On A^c , $h(x) = 0$, so it *does not matter* what $w(x)$ does there for the variance of this estimator.

Suppose we choose g such that it *oversamples* A :

$$g(x) > f(x) \quad \text{for } x \in A.$$

Then

$$w(x) = \frac{f(x)}{g(x)} < 1 \quad \text{for } x \in A,$$

hence $w(x) - 1 < 0$ on A , and

$$\Delta(h; f, g) = \int_A (w(x) - 1) f(x) dx < 0.$$

Therefore

$$\text{Var}_g(\hat{\mu}_{IS}) < \text{Var}_f(\hat{\mu}_{MC}).$$

Intuition. - Under f , the event A is rare; many MC runs produce zero or very few hits in A . - Under g , we *force* more samples to fall in A . - The weights correct for the change of distribution, but the fact that g focuses on the rare region drastically reduces the variance.

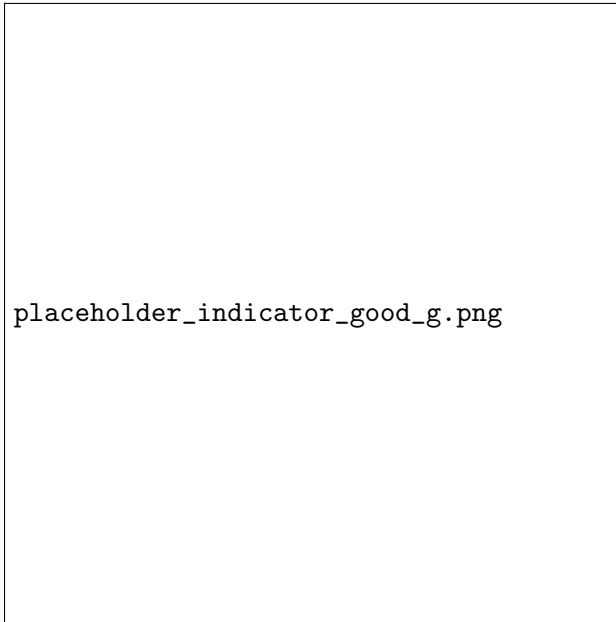


Figure placeholder. $f(x)$ and $g(x)$ drawn; A is a rare tail region. Here $g(x)$ is larger than $f(x)$ on A , so $w(x) < 1$ where $h^2(x) = 1$. Outside A , $h(x) = 0$, so the behavior of $w(x)$ on A^c is irrelevant for variance.

7.2 Example 2: Same indicator, but poor proposal

Now keep the same A and $h(x) = \mathbb{1}_A(x)$, but choose a proposal g that *undersamples* A :

$$g(x) < f(x) \quad \text{for } x \in A.$$

Then

$$w(x) = \frac{f(x)}{g(x)} > 1 \quad \text{on } A,$$

so

$$\Delta(h; f, g) = \int_A (w(x) - 1) f(x) dx > 0.$$

Hence

$$\text{Var}_g(\hat{\mu}_{IS}) > \text{Var}_f(\hat{\mu}_{MC}).$$

Intuition. - The event A is already rare under f . - If g makes it *even rarer*, then the few samples that land in A get very large weights $w(x) \gg 1$. - These few heavily-weighted samples create large fluctuations in the estimator and blow up the variance.

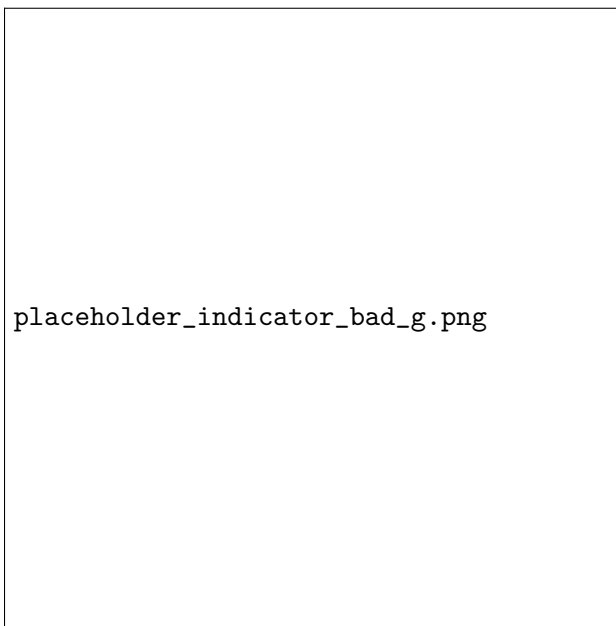


Figure placeholder. Same rare region A , but now $g(x) < f(x)$ on A , so $w(x) > 1$ in the only region where $h^2(x)$ is non-zero. Importance sampling makes variance worse than plain MC.

This shows that:

IS is not “automatically good” for rare events. It is only good if the proposal actually oversamples the rare event region.

7.3 Example 3: Smooth $h(x)$ (mean-type function)

Consider a smooth function $h(x)$ that is not concentrated on a tiny set; for example, $h(x) = x$ or $h(x) = x^2$ or some bounded smooth function.

Here, the “important” region is not a sharp set like A , but the set of x where $h^2(x)$ is large. The condition

$$\Delta(h; f, g) = \mathbb{E}_f[h^2(X)(w(X) - 1)]$$

says:

- if g oversamples (i.e. has $w(x) < 1$) in regions where $h^2(x)$ is large, IS can reduce variance; - if g undersamples those regions (i.e. has $w(x) > 1$ where $h^2(x)$ is large), IS makes variance worse.



Figure placeholder. $f(x)$ and $g(x)$ with a smooth $h(x)$. Where $h^2(x)$ is large, we want $g(x) > f(x)$ (i.e. $w(x) < 1$). If instead g focuses on regions where $h^2(x)$ is small, importance sampling wastes effort and adds noise.

This is the general rule:

For a fixed pair (f, g) , importance sampling is only good for those integrands h whose “energy” $h^2(x)$ is concentrated in regions where the proposal distribution g oversamples relative to f .

8 Summary

We considered two unbiased estimators of

$$\mu = \mathbb{E}_f[h(X)] :$$

- Plain Monte Carlo:

$$\hat{\mu}_{MC} = \frac{1}{n} \sum_{i=1}^n h(X_i), \quad X_i \sim f,$$

- Importance sampling:

$$\hat{\mu}_{IS} = \frac{1}{n} \sum_{i=1}^n h(X_i) \frac{f(X_i)}{g(X_i)}, \quad X_i \sim g.$$

Both are unbiased:

$$\mathbb{E}_f[\hat{\mu}_{MC}] = \mathbb{E}_g[\hat{\mu}_{IS}] = \mu.$$

Their variances differ by

$$\text{Var}_g(\hat{\mu}_{IS}) - \text{Var}_f(\hat{\mu}_{MC}) = \frac{1}{n} \mathbb{E}_f[h^2(X)(w(X) - 1)].$$

Thus:

- IS has *smaller* variance iff

$$\mathbb{E}_f[h^2(X)(w(X) - 1)] < 0,$$

i.e. $w(x) < 1$ on average where $h^2(x)$ is large (proposal oversamples the important region).

- IS has *larger* variance if this expectation is positive, i.e. $w(x) > 1$ where $h^2(x)$ is large (proposal undersamples the important region).

This gives a precise mathematical version of the intuition from the dice example: weights correct the *mean* automatically, but whether they reduce or increase *variance* depends entirely on where the proposal g puts its probability mass relative to $h^2(x)$.